

Akabir Abbas

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Karachi, Pakistan

Summary

AI/ML Engineer based in Karachi, Pakistan, with professional experience designing, developing, and deploying machine learning, deep learning, Generative AI, and LLM-powered solutions. Skilled in Python, PyTorch, TensorFlow, scikit-learn, FastAPI, LangChain, PostgreSQL, Docker, MLflow, DVC, and CI/CD, with experience in data preprocessing, feature engineering, model training, evaluation, deployment, monitoring, NLP, Computer Vision, REST APIs, and scalable AI systems. Available for onsite, hybrid, or remote roles and open to relocation for suitable opportunities.

Experience

Machine Learning Engineer

Jul 2026 – Present

Theta x Tech (SMC-PVT.) LTD, Karachi, Pakistan

- Design and develop Python-based AI/ML solutions, data preprocessing pipelines, predictive models, LLM applications, and REST APIs.
- Train, evaluate, optimize, and integrate models into scalable applications while improving accuracy, efficiency, and reliability.

AI Automation Engineer Intern

Nov 2025 – Mar 2026

Zorq Studio — Remote

- API integrations, data-processing pipelines, validation modules, and reporting systems.
- Worked with LLM-powered workflows, structured outputs, testing, debugging, logging, documentation, and software reliability improvements.

Education

Bachelor of Science in Computer Science

2022 – 2026

IBA University

Core Academic Subjects: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Software Engineering, Database Systems, Data Structures and Algorithms.

Technical Skills

Programming & Data: Python, SQL, pandas, NumPy, Data Cleaning, Data Preprocessing, Feature Engineering, ETL/ELT

Machine Learning: scikit-learn, Classification, Regression, Model Training, Model Evaluation, Hyperparameter Tuning, Performance Optimization

Deep Learning & Computer Vision: TensorFlow/Keras, PyTorch, Neural Networks, OpenCV, OCR, Image Processing

Generative AI & NLP: OpenAI API, GPT-4.1, LangChain, RAG Pipelines, Prompt Engineering, LLMs, AI Agents, Embeddings, NLP

MLOps & Monitoring: MLflow, DVC, Great Expectations, Experiment Tracking, Model Versioning, Drift Detection, Model Monitoring, Data Quality

Backend & Databases: FastAPI, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Alembic, Celery

DevOps & Cloud: Docker, Docker Compose, GitHub Actions CI/CD, Git/GitHub, Linux/WSL2, AWS/Azure Awareness

Projects

ModelOps Doctor

FastAPI, React, PostgreSQL, MLflow, DVC, Docker

- Built a full-stack MLOps platform with model registry, dataset profiling, prediction logs, classification metrics, drift detection, health scoring, and retraining recommendations.
- implemented Redis, Celery, CI/CD, monitoring, and structured logging.

Automated Bubble Sheet & Essay Evaluation System

Python, OpenCV, CNN, OCR, NLP, GPT-4.1

- Developed an AI evaluation system combining OMR grading, OCR extraction, CNN-based recognition, NLP assessment, GPT-assisted scoring, and automated reporting.
- Integrated Google Cloud Vision API and trained Computer Vision models, achieving **98.2% accuracy** on the final CNN model.

PromptVault

FastAPI, PostgreSQL, Redis, Celery, Airflow, Docker

- Built a production-style backend with REST APIs, PostgreSQL, Redis caching, Celery jobs, Docker, and GitHub Actions CI/CD.
- Added **61 automated tests** with approximately **63% coverage**, request tracing, and deployment-ready service architecture.

Certifications

Machine Learning Engineer Track (DataCamp) | Working with the OpenAI API (DataCamp) | Google AI Essentials (Google) | AI Skills Fest 2026 (Microsoft) | MLOps Concepts (DataCamp) | Introduction to MLflow (DataCamp)